

Examiner
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6. Consumers rely on a continuous supply of electricity from the National Grid system.

(a) Describe how the National Grid provides a reliable supply of electricity as efficiently as possible. [6 QER]

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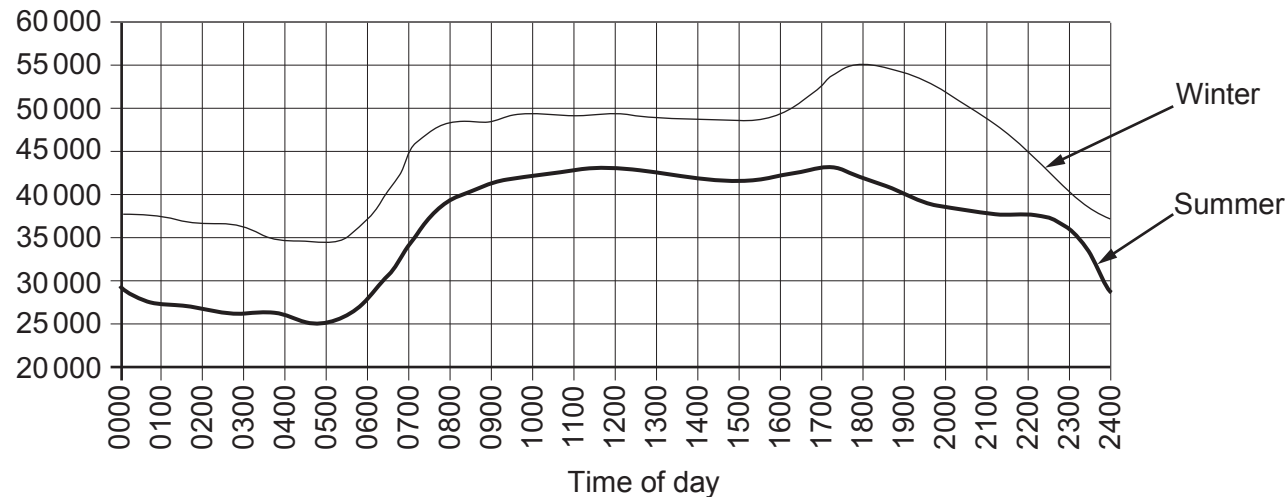
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(b) Information about the National Grid is shown below.

Typical summer and winter demands

National Grid demand (MW)



Network size

- Maximum demand 63 GW
- Output capacity 79.9 GW
- Annual mean electrical energy used in the UK 360 TWh
- Length of 400 kV grid 11 500 km
- Length of 275 kV grid 9800 km
- Length of 132 kV (or lower) grid 5 250 km

Energy losses at maximum demand

- Heating in cables 860 MW
- Fixed losses 260 MW
- Sub-station transformer heating losses 145 MW
- Generator transformer heating losses 155 MW

Total losses = 1 420 MW

(i) Use the information above and an equation from page 2 to calculate the % efficiency of the National Grid system at maximum demand. [3]

% efficiency =



(ii) Use the equation:

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

to calculate the annual mean power used in GW.
(1 year = 8 766 hours)

[3]

Mean power = GW

(iii) Tick (✓) the boxes next to the **two** correct statements about the data opposite. [2]

At maximum demand there is spare capacity of 16.9 GW ☐

Maximum demand is equivalent to 85% of total capacity ☐

The difference in minimum demand between summer and winter is approximately 10 000 MW ☐

The times of peak demand in summer and winter are **exactly** the same ☐

The difference in maximum demand between summer and winter is 15 000 MW ☐

END OF PAPER

